Shiv Modi

Software Engineer

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EDUCATION

Indian Institute of Technology Bombay, *Grade: 8.60/10.00* [2024]
Master of Technology, Computer Integrated Manufacturing
Bachelor of Technology, Mechanical Engineering, Minor in Industrial Engineering & Operations Research (IEOR)

Eklavya School of Science, *Grade (PCM): 89.50%*, Higher Secondary Certificate (Class XII), GHSEB [2018]

Adarsh Vidhyalaya, *Grade: 91.33%*, Secondary School Certificate (Class X), GSEB [2016]

SCHOLASTIC ACHIEVEMENTS & AWARDS

- Attained **AA grades** in **ME781 - Statistical ML & Data Mining** and **GNR638 - ML for Remote Sensing** courses [2022]
- Awarded the **Young Researchers' Prize** by **IEEE OES** by the **University of Tokyo, Japan** as **AUV-IITB** team member [2021]
- Obtained an All India Rank of **644** in **JEE Mains** (1.15 million appeared) | **97.9 percentile** in **IIT-JEE Advanced** [2019]
- Secured **1st rank** in Gujarat State with an All India Rank of **48** in **JEE-Mains Paper-2 (B Arch./B. Planning)** [2019]
- Obtained **70/70** in **Science** in SSC (X) (**99.74** percentile) and **100/100** score in **Mathematics** in HSC (XII) [2016-18]

PROFESSIONAL EXPERIENCE

Software Engineer | *Spectral Insights Pvt Ltd (Evident - Pramana Unit), Bangalore* *[Jul '24 - Present]*

AI-driven health-tech company specializing in digital pathology, building autonomous whole-slide imaging systems and software.

- **Camera IQC:** Enhanced camera quality assurance by integrating intensity-based and spatial clustering algorithms for defect classification, ensuring only high-reliability cameras are integrated into whole-slide imaging systems.
- **Hazy Edge Correction:** Eliminated **hazy stitching artifacts** in whole-slide images (WSI) by designing an overlap correction pipeline, improving image clarity across diverse slide types and scanner models.
- **Even Stack Acquisition:** Redesigned Z-stack acquisition to support both even and odd stack sizes, improving pipeline flexibility and delivering **3.25% faster scans per mm²** with **no loss** in image quality.
- **Rescan Optimization:** Improved rescan decision-making by filtering artifact-driven focus errors, achieving a **68.9% reduction** in unnecessary rescans and significantly enhancing tissue-focused image quality.

Data Analytics Intern | *MedTourEasy, New Delhi* *[May '23]*

*Blood Donations Prediction, Received **appreciation certificate** and **letter of recommendation***

- Developed an ML model using **logistic regression** for identifying patterns and factors influencing blood donation behavior
- Achieved an AUC score of **0.7634** by performing an intelligent search over ML pipelines to select an algorithm using **TPOT**
- Enhanced the AUC score by **3.37%** through the strategic application of **log normalization** to the high-variance features

Technical Intern | *Wings EV, Indore* *[Dec '21 - Jan '22]*

- Developed a **C++ library** on Raspberry Pi to process CAN bus data, enabling **audio alerts** based on received signal bytes
- Integrated functionality to play audio files through a speaker, enhancing **real-time notification** capabilities for the system

MASTERS' THESIS & OTHER KEY PROJECTS

AI Classifier for Wear and Lubrication State Identification *[Jun '23 - Jun '24]*

Received AB grade in M-Tech Project | Department of Mechanical Engineering, IIT Bombay *Guide: Prof. Asim Tewari*

- Automated **lubrication state monitoring** and **friction force prediction** in industrial machinery, reducing manual checks
- Designed custom **charge amplifier circuits** and **Arduino/ESP-32** systems for **high-frequency vibration** data acquisition
- Applied statistical ML models (RF, XGBoost, Logistic/Linear Regression) with feature engineering, PCA, and scaling
- Achieved **>98%** lubrication state classification accuracy and $\mathbf{R^2 = 0.83}$ in friction force prediction, reducing downtime

Virtual Moon Lander | *Machine Learning Specialization | Stanford University* *[Feb '23 - Jul '23]*

- Designed Rocket Trajectory Optimization **RL-Model** using **TensorFlow** and OpenAI's Environment Rendering Library Gym
- Reduced **network instabilities** and **oscillations** problems using **Experience Replay** and **Soft Weight Update** techniques
- Solved RL-Environment for absolute precision safety landing, achieving a **threshold score** of **200+** in just **534 Episodes**

Matsya, Autonomous Underwater Vehicle (AUV) *[Oct '19 - Jul '22]*

RoboSub, RoboNation *Project Guide: Prof. Leena Vachhani, Prof. Hemendra Arya*

*All student team working on the design & development of a state-of-the-art AUV, **Matsya**, which competes at **RoboSub, USA***

Accolades: IEEE Young Researchers' Prize awardee, University of Tokyo; 3rd prize in RoboSub 2020; Featured in Janes

Electrical Subdivision

- Designed **Power Board** in EAGLE which has **12 independent sensors** toggle lines, **reverse-voltage protection**, Hard-kill for thrusters, the ability to generate required multiple voltages across the vehicle and measure the voltage of a battery
- Developed the code for the microcontroller performing functions like **I/O signals**, Power toggles, **UART** communication

TECHNICAL SKILLS & CERTIFICATIONS

C++, Python, R, SQL, MongoDB, Keras, Scikit-Learn, Pandas, NumPy, SciPy, Matplotlib, Seaborn, Plotly, Tableau, MS Excel, IBM Cognos, MATLAB, Git/GitHub | **Certifications:** Google Data Analytics, IBM Data Science, Stanford ML Specialization